

Rosen — §6.1: The Basics of Counting

Selected Exercises

Alston

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Exercise 39

How many partial functions (see Definition 13 of Section 2.3) are there from a set with m elements to a set with n elements, where m and n are positive integers?

Exercise 41

A palindrome is a string whose reversal is identical to the string. How many bit strings of length n are palindromes?

Exercise 50

How many bit strings of length 10 contain either five consecutive 0s or five consecutive 1s?

Exercise 51

How many bit strings of length eight contain either three consecutive 0s or four consecutive 1s?

Exercise 70

Use the product rule to show that there are 2^{2^n} different truth tables for propositions in n variables.